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A Serious Game to Promote Water–Energy–Land–Food–People (WELFP) Nexus Perception and Encourage Pro-Environmental and Pro-Social Urban Agriculture

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Abstract: Urban agriculture is key to sustainable city development, particularly through public engagement with the Water–Energy–Land–Food–People (WELFP) Nexus. This study examines the effectiveness of serious games in enhancing WELFP understanding and promoting pro-environmental and pro-social behaviors. A game-based learning model was developed using the Stimulus–Organism–Response (SOR) and Easy–Attractive–Social–Timely (EAST) frameworks, along with the Revised New Ecological Paradigm (NEP) Scale. The model simulates real-world urban agriculture challenges to foster participatory decision-making. A survey of 200 urban agriculture practitioners, analyzed via structural equation modeling (SmartPLS 4.0), found that perceived timeliness (PT) and perceived usefulness (PU) significantly influenced both the perceived sustainable livelihood value (PT: $p = 0.000$; PU: $p = 0.006$) and users' attitudes toward the game (PT: $p = 0.000$; PU: $p = 0.038$). While enjoyment positively affected attitude ($p = 0.002$), it negatively impacted perceived value ($p = 0.002$), revealing a trade-off between fun and practical relevance. Perceived ease of use improved perceived value ($p = 0.000$) but did not affect attitude, suggesting emotional engagement matters more. Both attitude and perceived value strongly predicted users' intention to engage with the game. Post-game reflections highlighted the need for cross-sector collaboration, strategic resource use, access to real-time data, and responsive crisis management. Participants also stressed the importance of public awareness, civic responsibility, and volunteerism in advancing community-driven sustainable agriculture. These findings highlight the need to balance engagement and educational depth in game-based learning for sustainability.

Keywords: WELFP nexus; pro-environmental behavior; pro-social behavior; urban agriculture; serious games; educational tool

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1. Introduction

Serious games provide an innovative approach to enhancing public understanding by immersing users in interactive decision-making scenarios [1,2]. These games simplify complex sustainability challenges, fostering informed, pro-environmental choices. Serious games have proven effective in raising environmental awareness, improving decision-making, and encouraging behavioral change [3]. However, most lack an integrated framework linking the Water–Energy–Land–Food–People (WELFP) Nexus components to urban agriculture and are often disconnected from real-world initiatives [4].

A systematic review underscores the need for diverse games, interdisciplinary integration, and behavioral impact assessment [3], along with evaluating serious games'

role in system dynamics and human–system interactions [2]. While serious games and gamified mobile apps raise awareness of environmental issues, their impact on real-world pro-environmental behavior remains uncertain, requiring research on bridging knowledge and action [5]. Moreover, research on their long-term impact and ethical concerns remains limited [6].

This study aims to develop, evaluate, and implement serious games to promote WELFP understanding and encourage pro-environmental and pro-social urban agriculture by accomplishing the following: (1) developing serious games that effectively communicate WELFP interconnections and ensure user engagement; (2) examining the factors influencing behavioral intention (BI) to apply and adopt what was learned from the game in real life, using a questionnaire survey and game debriefing; and (3) proposing insights for policy development based on player feedback and game outputs. The innovation of this game, compared to existing WELFP educational tools, lies in its enhanced behavioral intervention mechanisms, which link nexus components to urban agriculture and are grounded in real-world initiatives. Furthermore, the game process incorporates an evaluation of serious games' role in system dynamics and human–system interactions, along with attention to pro-environmental and pro-social concerns. By addressing these gaps, this research seeks to advance serious game development and promote real-world sustainability transitions in urban agriculture.

2. Materials and Methods

2.1. Developing Serious Game "NEXUS Farming: Balancing Resources for Well-Being"

2.1.1. Game Design

The game was adopted from the previous NEXUS game by the Centre for Systems Solutions [7]. Moreover, the game was designed to provide the opportunity for players to realize the real situations of ecological crisis and the limits of nature, including the capacity of humans to manage and balance human well-being and the capacity of nature, which was based on the revised NEP Scale [8]. The game also motivates the players to reflect their pro-environmental and pro-social behaviors in the urban agricultural context.

Pro-environmental behavior was designed based on three types of positive pro-environmental behavior [9]: (1) contributing to nature (CTN), (2) contributing to science and practical solutions (CTS), and (3) contributing to useful information (CTI). As noted in reference [10], pro-social behavior is collectively termed contributing to people (CTP) in this study (Table 1).

After drafting the game, a consultation process was carried out with three experts in educational game development, urban agriculture, and social science. The game was then pre-tested twice with a small group of five urban farmers to ensure their understanding of the game process and its alignment with the intended goals. Subsequently, the final content of the role and situation cards was refined and designed for inclusion in the game (Table 2, Figure S1).

Table 1. Ideas to develop the game and the games design.

Ideas to Develop the Game	Game Design
Revised NEP Scale [9]: ecological crisis, limits of nature, capacity of humans to manage and balance	The current environmental and natural resource situation in communities, both nationwide and worldwide, is addressed. Crisis conditions during the game include floods and droughts. Resources can be shared with other players to manage these crises, as explained in the first step of the game process.
Pro-environmental [9] and **pro-social** [10] **behaviors**	Players are required to:

Table 1. *Cont.*

Ideas to Develop the Game	Game Design
1. Contribute to nature (CTN) by engaging in activities that support and protect the environment.	- Concern about the interlink among the natural resources
2. Contribute to science and practical solutions (CTS) by promoting scientific knowledge and practical solutions.	- Think about the solution that should be appropriate with their community context.
3. Contribute to useful information (CTI) by sharing experiences and insights on relevant issues.	- Share their experiences or knowledge based on the issues discussed.
4. Contribute to people (CTP) by expressing empathy, helping, sharing, and volunteering.	Players have the opportunities to reflect their empathy, help, share and volunteer or contribution with other players based on the situation cards of flood and drought for collaborative solving.

Note: adjusted from Discovery: the players are doing discovery activities.

Table 2. The role and situation cards.

Role Cards		Situation Cards			
		Situations	Impact	Mission	Outcome
1. Provincial Irrigation Project Director	Manage water resources equitably for all sectors (drinking water, domestic use, and ecological balance).	Floods and droughts cause damage to various sectors: agricultural crops, food supply, roads and infrastructure, electricity supply, tap water system, buildings and houses, and tourism industry.	Insufficient clean water to meet the needs of all sectors.	Find ways to allocate water adequately for each sector.	1. Allocate resources to all sectors without causing conflicts = 🍽🍽 + ☺☺ 2. Most players agree with the management = 🍽 + ☺ 3. Cannot make the final decision together = ○ + ☹
2. Provincial Electricity Authority (PEA)	Generate renewable energy to meet the demands of all sectors.		Energy production must temporarily halt for inspection/repair before returning to normal operation.	Find ways to allocate energy adequately for each sector.	
3. Subdistrict Agricultural Officer	Promote environmentally friendly agricultural practices among farmers and interested individuals.		Farmers face production losses, leading to food shortages in the community.	Promote knowledge and skills for restoring vegetable plots after floods and droughts and encourage household vegetable gardening.	
4. Owner of a Vegetable Farm Resort	- Owns land for vegetable cultivation to serve tourists and supply local markets. - Provides accommodation and food services for visitors.		Vegetable plots, soil, and crops are damaged. Lack of tourists leads to reduced income, requiring staff layoffs.	Restore vegetable plots and prepare for welcoming tourists.	
5. Community Thought Leader	- Ensure the well-being of community members. - Act as a liaison between the community and other sectors.		The community faces food shortages.	Encourage community members to grow household vegetables for food security. However, some people lack planting space, have limited time, or live in low-light areas.	

2.1.2. Game Process

Participants' Selection

In this study, 200 urban agricultural farmers from 10 areas in Bangkok, Pathumthani, Chonburi, Nakhon Ratchasima, and Songkhla, were recruited to play the game. Each area consisted of four groups, and each group included five players who played the game simultaneously. The main qualification for participant selection was having experience related to urban agriculture in their communities. This ensured that participants had sufficient background knowledge to meaningfully engage with the game content and provide

informed feedback. Furthermore, their experience enabled a more accurate evaluation of how well the game reflected real-life urban agricultural practices. Participants were available and willing to engage in the game sessions for approximately two hours.

Process of the Game Play

The process of playing the game was gradually explained while providing the relevant cards (Figure S1) to the players. The researcher, acting as the game facilitator, explained the gameplay process as described below and summarized in Figure 1:

(1) The current situation of humanity and nature are as follows: the Earth has limited resources and space, yet human populations continue to grow. While some believe we have the right to modify nature to meet our needs, others argue that interference often leads to environmental destruction. Despite human ingenuity, reckless exploitation of natural resources and industrial impacts are pushing ecosystems to their limits. Nature's balance is both resilient and fragile—strong enough to withstand some changes yet easily disrupted by overuse and pollution. Though humans possess advanced knowledge, we remain subject to nature's laws. If we fail to manage resources wisely, an ecological crisis could become unavoidable. The future depends on whether we learn to coexist with nature or attempt to dominate and control it. Will innovation save us, or will unsustainable actions lead to disaster? The choice is ours;
(2) All players were requested to select their own role. They can consult together which role for each is more appropriate and convenient. They will be informed that during the games, they need to solve the problems individually and all together in a group;
(3) All players received a role card as their selection, and 5 Water Tokens and 5 Energy Tokens. Then, they have 5–10 min to read and understand the assigned role.

Knowledge cards or assistance cards (Figure S2) were also provided as example guidelines for problem solving covering three main groups consisting of water (six example cards), energy (seven example cards), and agricultural (twelve example cards) managements (as shown more in the Supplementary Materials);

(4) During the normal conditions: each sector needs to use the natural resources as follows: (1) household/office/farm resort: 2 water tokens, 2 energy tokens, and 1 pollution token; and (2) water allocation for ecological balance: 2 water tokens and 2 energy tokens;
(5) During the crisis conditions (flood/drought): water and energy tokens can be shared with other players to benefit the community as a whole. To solve flood and drought situations, one mission needs 2 water tokens and 2 energy tokens;
(6) However, if the renewable energy or environmentally friendly agricultural practices cannot be adopted, 1 pollution token will occur for each situation. If the community has 4 pollution tokens, meaning that it is exceeding the limit of the nature, 1 water token or 1 energy token is needed to purify the pollution;
(7) Solution criteria should benefit the majority or whole society and minimize negative impacts on all groups;
(8) Group discussion: collaborate on solutions for each member's event within around 15 min; so, in total for 5 players, a mission will take around 1.15 h;
(9) If the group discussion came up with the resources allocation to all sectors without any conflicts, the group will receive two healthy food cards and two happiness cards. If only most players agree with the solution, only one healthy food card and one happiness card would be gained. However, if the players cannot make the final decision together, the group will receive no food and no happiness card;
(10) The game ends when all players have discussed and resolved every situation in their group;

(11) After completing the games, all five players in each group were asked to reflect on their learning and plan for future practice through a focus group discussion lasting about 20–30 min. During the debriefing step, the open-ended question "What have you learned from engaging in the games?" was asked to elicit more details about their feelings and experiences from participating in the game.

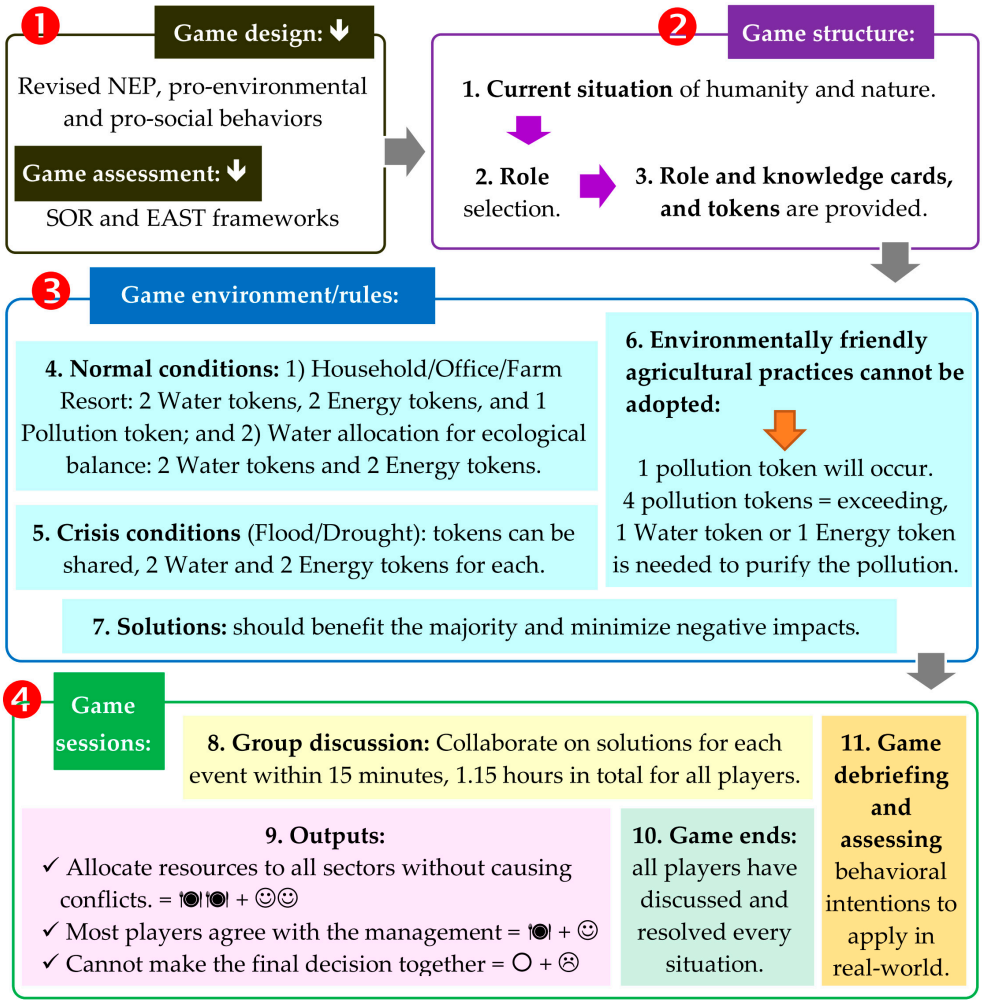


Figure 1. Process of the game play.

2.2. Examining Factors Influencing the Behavioral Intention to Apply and Adopt What Was Learned from the Game in Real Life

This study employed revised NEP [8] and pro-social [10] concepts to develop serious games that effectively communicate WELFP interconnections and ensure user engagement. Furthermore, to examine the links between WELFP perception, moral intuitions, and sustainable livelihoods, the SOR [11] and EAST [12] frameworks were adopted to reflect the players learning and intention to applying in the real world.

The SOR framework has been widely used to explain consumer behavior and decision-making processes [13]. The model has three key components: stimuli (S), the external factors affecting the user; organism (O), the psychological process of interpreting those stimuli; and response (R), the user's behavioral reaction to the stimuli [14]. More recently, it has been applied to pro-environmental behavior to understand how external environmental factors (stimuli) influence individuals' internal psychological states (organism), ultimately shaping their eco-friendly actions (response) [15]. A major strength of this framework lies in its flexibility, allowing researchers to explore a broad range of stimuli—tangible and intangible

and internal and external, including experiential and non-experiential elements—as well as various types of responses [13].

Meanwhile, the EAST framework [12] focuses on how players perceive the game in terms of ease, attractiveness, social relevance, and timeliness. For example, assigning players a role in the game makes the learning experience more personal and encourages them to reflect on timely issues in their own lives.

The integration of the SOR and EAST frameworks strengthens the connection to real-world intentions. The SOR model explains why players respond to game stimuli, while EAST informs how to design those stimuli to maximize impact. Through internal processing (the "organism" in SOR), players gain insights and form intentions, especially when the experience is easy to understand and timely (as emphasized in EAST). The final "response" may be a real-world action, such as starting a home garden or discussing composting with family members. Therefore, the combined strengths of the SOR and EAST frameworks were harnessed to effectively reflect players' learning and their intention to apply it in real-world contexts.

For this study, the SOR framework (Figure 2) was adopted as follows: the stimulus consisted of (1) perceived ease of use (PEOU) or easy; (2) perceived usefulness (PU) or attractive; (3) perceived enjoyment (PE) or attractive; (4) perceived practicality (PP) or attractive; (5) perceived social norm (PSN) or social; and (6) perceived timeliness (PT) or timely. Organism covered both attitude (ATT) and perceived value (PV) in terms of social, human, physical, financial, and natural benefits, while response focused on behavioral intention (BI).

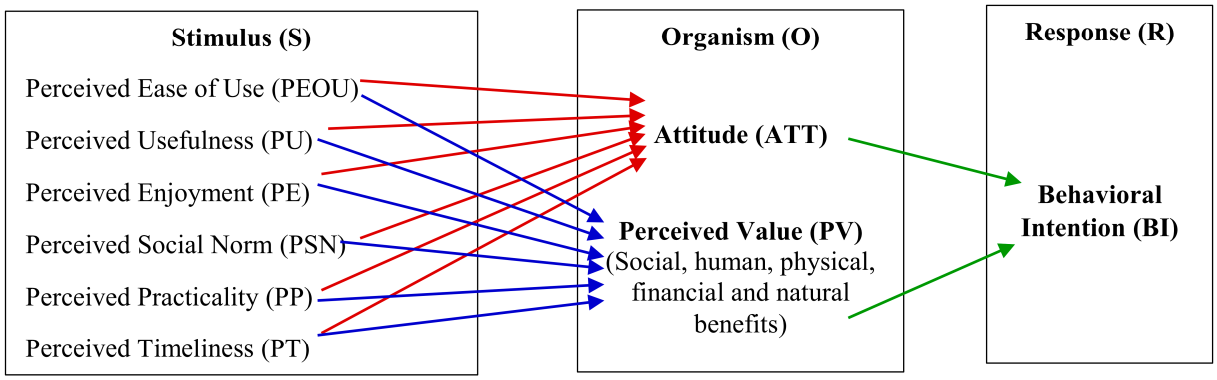


Figure 2. Conceptual framework.

The SOR and EAST frameworks were applied to frame the questions for the questionnaire survey to explore the linkage between WELFP perception and pro-environmental and pro-social urban agriculture (Table 3). These frameworks help expand and deepen the dimensions of the questions following gameplay, reflecting players' learning and their intentions to apply this knowledge in real life.

Table 3. Constructs and measurement items.

ELM Components	Questions	Sources
1. Perceived ease of use (PEOU)	PEOU1: Learning to use the game is easy for me. PEOU2: Gaming process is easy to use. PEOU3: The interaction during the game is required less mental effort.	[12,16]
2. Perceived usefulness (PU)	PU1: This game increases effectiveness of UA resources nexus perception. PU2: This game increases effectiveness of UA practice. PU3: This game increases effectiveness of UA policy.	[12,17]
3. Perceived enjoyment (PE)	PE1: I think that this game is enjoyable. PE2: I think that this game is interesting. PE3: I think that this game is exciting.	[12,18]

Table 3. *Cont.*

ELM Components	Questions	Sources
4. Perceived social norms (PSN)	PSN1: This game help learn others' sustainable behavior. PSN2: This game promote the group identity. PSN3: This game promote social connections.	[12]
5. Perceived practicality (PP)	PP1: This game is set with practical situation for players to engage. PP2: This game process is set logically for players to engage. PP3: The roles of all players in this game is appropriate to apply in the real practice.	[12]
6. Perceived timeliness (PT)	PT1: This game help encourage pre-commitments and emphasize present benefits. PT2: This game help create timely moments. PT3: This game help plan and follow through in UA practice.	[12]
7. Attitude (ATT)	ATT1: For me, this game is very good. ATT2: For me, this game is important. ATT3: I like the idea of this game.	[19]
8. Perceived value (PV)	PV1: This game is social beneficial to me. PV2: This game is human beneficial to me. PV3: This game is physical beneficial to me. PV4: This game is financial beneficial to me. PV5: This game is natural beneficial to me	[20]
9. Behavioral intention (BI)	BI1: I intend to apply what I have learned from playing the game in the real life. BI2: I predict that I would adopt what I have learned from playing the game in the real life.	[17]

2.3. Data Analysis

Running a factor analysis in Smart PLS is part of the process of assessing the measurement model by checking outer loadings, Cronbach's alpha, Average Variance Extracted (AVE), Composite Reliability, and discriminant validity to explore the underlying factor structure of observed variables or indicators. All the outputs were acceptable in this study as explained in Section 3.

Qualitative data were analyzed using QDA Miner Lite 4.0 for coding, labeling, and grouping. Results were refined through researcher discussions and independent reviewer feedback. Cohen's Kappa indicated significant agreement ($p < 0.001$). To ensure reflexivity, we maintained detailed records and sought critical feedback, drawing on our expertise in communication studies, environmental science, and agricultural communication.

To ensure trustworthiness, we followed Lincoln and Guba's framework [21] using methodological triangulation for credibility, detailed audit trails for dependability, peer debriefing for confirmability, and purposive sampling for transferability.

3. Results and Discussions

Factors influencing the behavioral intention to apply and adopt what was learned from the game in real life were analyzed using Smart PLS 4.0. Both the measurement model (validity and reliability) and the structural model (hypothesized relationships) were tested, with the results as follows.

3.1. Measurement Model

In this study, the measurement model confirmed convergent and discriminant validity. The item loadings, Composite Reliability (CR) and Cronbach's alpha were greater than 0.70, and the AVE was greater than 0.50 (Table 4), indicating reliability and convergent validity [22], while the HTMT was found to be less than 0.85 (Table S2), confirming the discriminant validity. Discriminant validity, ensuring that the square root of the AVE for each construct is greater than the correlations between the constructs [23], was also achieved (Table S3).

Table 4. Factor loadings, reliability, and validity of construct items (sample N = 200).

Variable	Measurement Item	Factor Loadings	Cronbach's Alpha (More than 0.7)	rho_a (More than 0.7)	CR or rho_c (More than 0.7)	AVE (More than 0.5)
Attitude (ATT)	ATT1	0.781	0.852	0.744	0.764	0.658
	ATT2	0.838				
	ATT3	0.813				
Behavioral Intention (BI)	BI1	0.920	0.877	0.725	0.772	0.781
	BI2	0.846				
Perceived Enjoyment (PE)	PE1	0.794	0.854	0.744	0.757	0.662
	PE2	0.870				
	PE3	0.772				
Perceived Ease of Use (PEOU)	PEOU1	0.763	0.827	0.704	0.717	0.614
	PEOU2	0.815				
	PEOU3	0.772				
Perceived Practicality (PP)	PP1	0.759	0.847	0.731	0.750	0.649
	PP2	0.803				
	PP3	0.852				
Perceived Social Norm (PSN)	PSN1	0.767	0.869	0.774	0.782	0.690
	PSN2	0.859				
	PSN3	0.863				
Perceived Timeliness (PT)	PT1	0.744	0.834	0.703	0.712	0.626
	PT2	0.806				
	PT3	0.822				
Perceived Usefulness (PU)	PU1	0.873	0.839	0.717	0.752	0.637
	PU2	0.742				
	PU3	0.773				
Perceived Values (PV)	PV1	0.728	0.856	0.791	0.793	0.544
	PV2	0.757				
	PV3	0.734				
	PV4	0.761				
	PV5	0.705				

3.2. Structural Model

In the structural model analysis (Table 5), the Variance Internal Factor (VIF) (Table S4), effect size (f^2), Coefficient of Determination (R2), Q^2, *t*-value, standard deviation, and coefficient value were analyzed. The VIF is used to assess multicollinearity among predictor variables in a structural model. This study found that the VIFs were 1.309–1.865, which were less than 3.0, which is acceptable using the common threshold and confirms no multicollinearity [24].

Table 5. The analysis of direct effects, total effects, and hypothesis testing.

Relationship	Path Co-Efficient	Std. Dev.	*t*-Value	Confidence Interval		f^2	VIF	*p* Values	Supported
				2.5%	97.5%				
PEOU → PV	0.304	0.054	5.641	0.194	0.406	0.127	1.297	0.000	Yes
PT → PV	0.409	0.063	6.505	0.283	0.532	0.196	1.520	0.000	Yes
PT → ATT	0.456	0.067	6.839	0.324	0.579	0.243	1.520	0.000	Yes
ATT → BI	0.358	0.086	4.154	0.181	0.518	0.147	1.164	0.000	Yes
PE → ATT	0.229	0.075	3.063	0.082	0.374	0.045	2.053	0.002	Yes
PE → PV	−0.251	0.080	3.131	−0.408	−0.095	0.055	2.053	0.002	Yes
PV → BI	0.245	0.082	2.998	0.090	0.405	0.069	1.164	0.003	Yes
PU → PV	0.167	0.061	2.747	0.050	0.290	0.035	1.414	0.006	Yes
PU → ATT	0.167	0.080	2.074	−0.002	0.317	0.035	1.414	0.038	Yes
PEOU → ATT	0.056	0.077	0.726	−0.092	0.208	0.004	1.297	0.468	No
PP → ATT	0.055	0.100	0.549	−0.142	0.255	0.003	2.113	0.583	No
PP → PV	0.118	0.104	1.133	−0.083	0.322	0.012	2.113	0.257	No
PSN → ATT	0.071	0.084	0.841	−0.109	0.226	0.004	2.097	0.400	No
PSN → PV	−0.125	0.079	1.578	−0.288	0.024	0.013	2.097	0.115	No

The analysis of direct and total effects (Table 5) involves calculating the path coefficients (ß), which should be ≥0.10 and statistically significant at the 0.05 level, to test the hypotheses. Hypothesis testing checks whether the path coefficients in the inner model are nonzero, signifying that the variable at the arrow's origin affects the one at its endpoint. Additionally, it is essential to confirm that the outer loading coefficients are nonzero.

The f^2 value measures the impact of an independent variable on a dependent variable. An f^2 value above 0.02 affects predictive accuracy, with thresholds set as follows: ≥0.02

for a low effect, ≥0.15 for a medium effect, and ≥0.35 for a high effect. In this study, effect sizes range from 0.003 to 0.243. PT to ATT (0.243) and PV (0.196) show medium effects, while the others remain low. Meanwhile, PE and PU affect ATT; PE, PEOU, and PU affect PV; and ATT and PV affect BI, all with low effects.

The R^2 values for ATT, PV, and BI are 0.437, 0.439, and 0.254, respectively, showing that PT, PE, and PU account for 43.7% of the variance in ATT. PEOU, PT, PE, and PU can explain 43.9% of the variation in PV, while ATT and PV can explain 25.4% of the variation in BI (Table S5).

The PLS predict MV summary shows the predictive performance of individual indicators that measure observed variables, while the PLS predict LV summary aggregates the predictive performance of all indicators related to a particular latent variable. The PLS predict LV summary reflected a high level for both ATT and PV, while a medium level was found for BI (Table 6 and Figure 3).

Table 6. The Q^2 prediction results and interpretation.

	PLS Predict MV Summary	Interpretation	PLS Predict LV Summary	Interpretation
ATT1	0.188	medium	0.381	high
ATT2	0.354	high		
ATT3	0.178	medium		
BI1	0.218	medium	0.180	medium
BI2	0.045	low		
PV1	0.203	medium	0.391	high
PV2	0.264	medium		
PV3	0.129	medium		
PV4	0.199	medium		
PV5	0.244	medium		

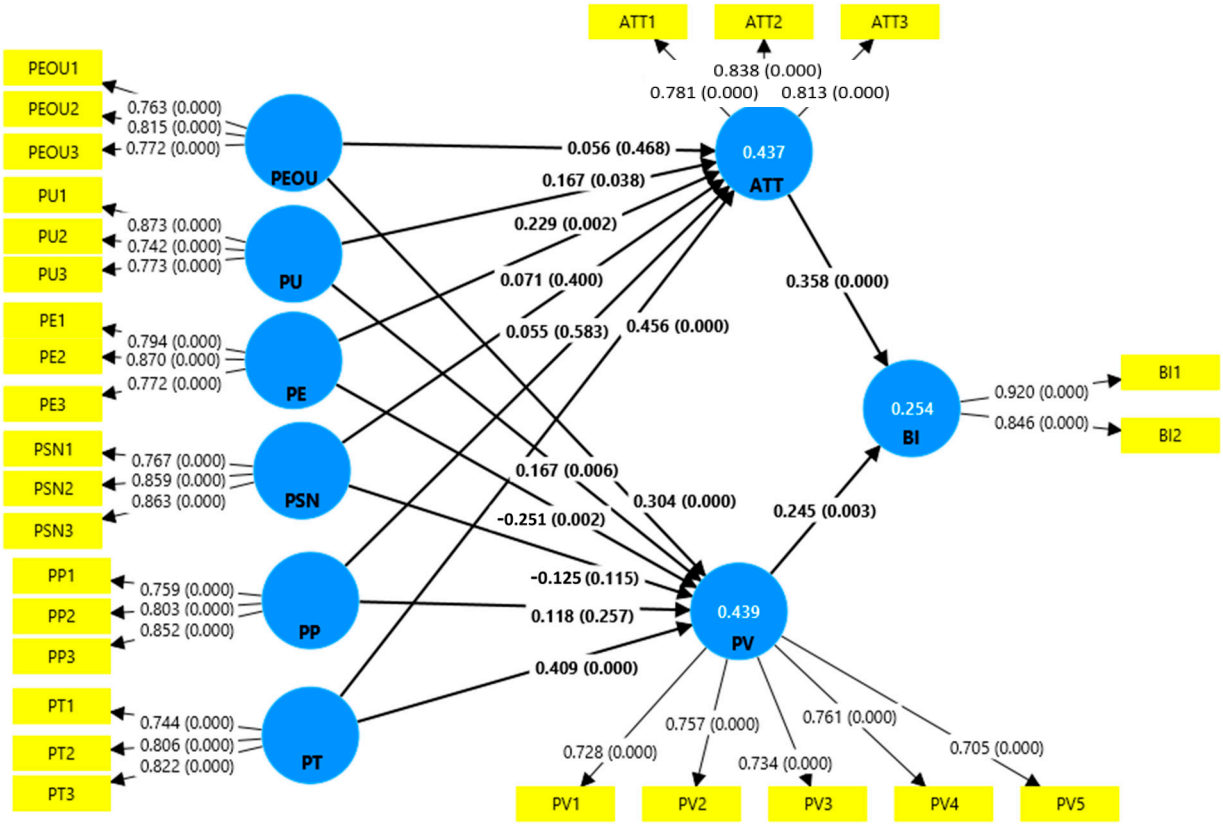


Figure 3. PLS analysis of key effects, including path coefficients, *p*-values, and R^2 values.

3.3. Perceived Value and Behavioral Intention: Players' Reflections and Discussions

Our analysis revealed that the perceived value (PV) of the game—evaluated across five dimensions of livelihood—was significantly influenced by four key factors: perceived ease of use (PEOU; $p = 0.000$), perceived timeliness (PT; $p = 0.000$), perceived enjoyment (PE; $p = 0.002$), and perceived usefulness (PU; $p = 0.006$). Perceived ease of use (PEOU) significantly enhanced the perceived value of the game ($p = 0.000$), yet it did not notably influence users' attitudes. In contrast, perceived enjoyment (PE) boosted users' attitudes but negatively affected how they viewed the game's practical usefulness. Both attitude (ATT; $p = 0.000$) and perceived value (PV; $p = 0.003$) significantly predicted behavioral intention (BI), specifically in terms of participants' willingness to apply and adopt lessons from the game in real-life contexts.

3.3.1. Players' Reflections

Participants reported that the game was easy to use and understand (PEOU). They emphasized that the game instructions were clearly delivered through both oral and written formats, making the learning process accessible (PEOU1). The structure of the game—organized into a seamless flow from introduction to gameplay and debriefing—was perceived as intuitive (PEOU2). They appreciated how tasks were broken down into manageable steps, reducing cognitive load. Participants also noted that support materials such as knowledge and assistance cards provided useful guidance during decision-making phases. The design encouraged natural, low-pressure discussions, requiring less mental effort (PEOU3), and the use of tokens and visual cues helped reinforce key concepts.

Regarding perceived usefulness (PU), players indicated that the game effectively illustrated the interconnections within the WEFLP nexus (PU1) and helped them understand the complexity of urban agriculture systems. They experienced simulated challenges like drought or flooding and explored practical responses through collaborative decision-making. Players mentioned that these scenarios mirrored real-world situations and reinforced the need for sustainable practices (PU2). Furthermore, stakeholder engagement involved taking on the roles of local policymakers for various types of natural resources (water, energy, and agriculture) and local leaders to understand policy challenges and trade-offs (PU3).

The perceived timeliness (PT) of the game was also well received. Participants noted that they were encouraged to make prompt decisions in response to dynamic events (PT2), such as sudden environmental changes or policy shifts. The immediate visual feedback—through changes in resources, pollution levels, and community well-being—helped them see the real-time consequences of their decisions (PT1). Players also reported that the game inspired long-term thinking and strategic planning (PT3), which could translate into actionable steps in their own communities.

The element of enjoyment (PE) further strengthened participants' engagement. They found the game fun and stimulating (PE1), especially due to its team-based structure and role-based gameplay. Players appreciated the opportunity to explore diverse perspectives through assigned roles (e.g., water manager, urban farmer, and local policymaker), and the use of realistic, localized scenarios made the experience feel relevant and exciting (PE2 and PE3). Unpredictable events and collaborative challenges created a sense of urgency and excitement, enhancing the learning experience. The findings are summarized in Figure 4.

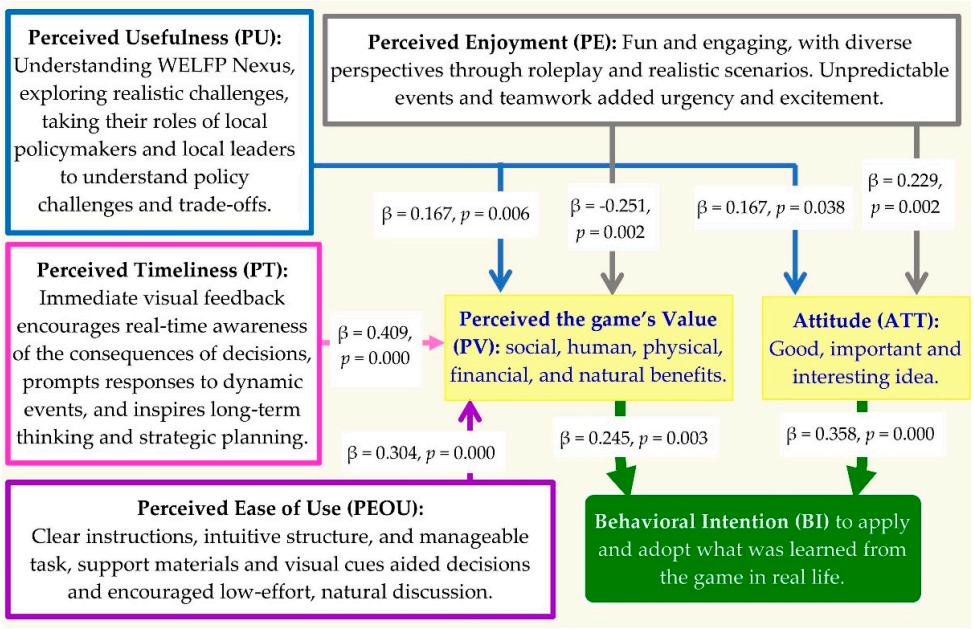


Figure 4. Factors influencing intent to apply the game's learning in real life.

3.3.2. Discussions from Players' Reflections

From a theoretical perspective, the findings align with the Technology Acceptance Model (TAM), which posits that users are more likely to adopt a system they perceive as both easy to use and useful [25,26]. In this study, perceived usefulness (PU) was clearly demonstrated, as players actively applied game strategies to address real-life urban agriculture challenges, such as managing droughts and floods. The game's embedded rewards and mini-challenges, based on sustainable practices like water efficiency or renewable energy use, acted as motivational reinforcements. These gamification elements, such as feedback loops and points, have been shown to enhance pro-environmental learning outcomes and player engagement in serious games [27,28].

Perceived ease of use (PEOU), while significantly improving perceived value, did not influence user attitudes toward the game. This suggests that although an intuitive interface supports cognitive engagement, it is not enough to generate positive affective responses. In serious games focused on sustainability, users' attitudes are often shaped more by emotional engagement and the relevance of content than by usability alone [26,29]. Moreover, in a purpose-driven context, such as among urban agriculture practitioners, ease of use becomes less critical, as users tend to prioritize meaningful content and outcomes over interface simplicity [30].

The perceived timeliness of information within the game supports the findings [31,32] that timely feedback fosters real-world applicability and adaptive thinking. Moreover, PE plays a critical role in influencing attitudes toward sustainable technology adoption [33–35].

The perceived value of the game was closely tied to players' sense of how urban agriculture could enhance their livelihoods. Participants viewed urban farming as a means to improve food security, reduce household costs, and increase local resilience. This perception significantly influenced their behavioral intention to adopt urban farming practices. Perceived livelihood value enhances behavioral intention by fostering positive attitudes and a sense of behavioral control, while heightened attention to urban agriculture—stimulated by the game further reinforces intention through increased awareness and interest [32,33].

Perceived enjoyment refers to the extent to which playing the game is perceived as pleasurable, fun, and intrinsically rewarding. In the context of this serious game, enjoyment strongly enhances users' affective evaluation or attitude toward the game. According to the TAM and its

extensions, enjoyment is a key predictor of attitude because it reinforces intrinsic motivation and a favorable emotional response [26,36]. When participants enjoy the NEXUS game, they are more likely to perceive it as likable, good, and important—regardless of whether they view it as practically useful for real-life livelihood decisions.

While enjoyment boosts affective responses, it can simultaneously undermine perceptions of practical utility, especially in educational or development-focused contexts such as sustainable livelihoods. This contradiction is explained by the "fun vs. seriousness paradox" [37]. If a game is perceived as too fun or entertaining, users may not take the content seriously. This is particularly relevant for topics like sustainable agriculture, income stability, and environmental management, which are complex, sensitive, and require realism. Participants might interpret the game's enjoyable nature as oversimplifying or trivializing serious issues, thereby reducing their perception of its relevance to real-life livelihood challenges.

In summary, players found the game to be accessible, informative, timely, enjoyable, and relevant. These perceptions, supported by established theoretical frameworks, underscore the potential of serious games like "NEXUS farming" to foster pro-environmental and pro-social behaviors in urban agriculture contexts.

3.4. The Key Messages Shared by the Players and Reflections on the NEXUS

The key messages emerged from the players' actual discussions and reflections during gameplay and were later categorized based on thematic analysis. They can be summarized into four main aspects of pro-environmental and pro-social behaviors, which include the following: (1) contribute to nature (CTN): support and protect nature and harm nature (HN); (2) contribute to science and practical solutions (CTS): promote science knowledge and activities including practical solutions; (3) contribute to useful information (CTI): share their experiences or knowledge; and (4) contribute to people (CTP): demonstrate empathy, volunteer or contribution, offer help, and share good things and opportunities (Figure 5).

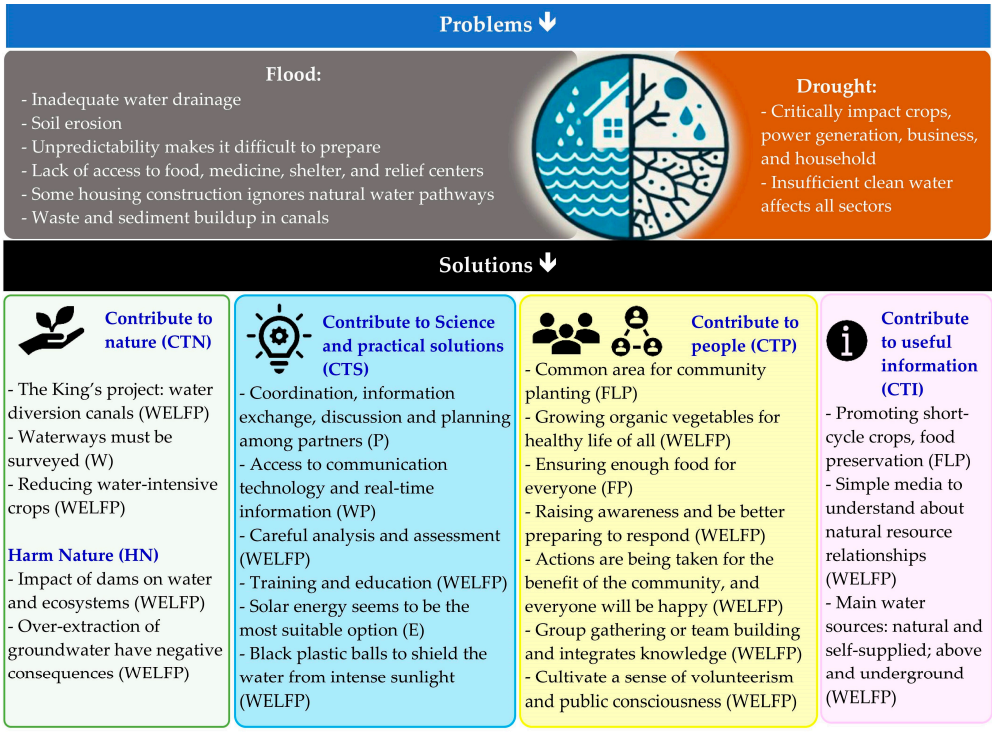


Figure 5. Key messages shared by the players and reflections. (Note: W: water; E: energy; L: land; F: food; and P: people).

3.5. Game Debriefing: Lessons-Learned by the NEXUS Game Players

Participants played the game primarily to learn about the interconnections between urban agriculture, environmental sustainability, and pro-social behaviors. While assessing the game and measuring the knowledge were secondary objectives, the main goal was to use the game as an educational tool to engage players and promote understanding of the WELFP Nexus. After playing the games, the players reflected on their learning and how it could be applied to their urban agriculture practices, which can be highlighted as follows:

Through the NEXUS game, players gained valuable insights into effective resource management, community collaboration, and sustainable problem solving. A key lesson was the importance of root cause analysis, where understanding the local context and past experiences of floods and droughts helps foster ongoing discussions and collective problem solving. Recognizing that water management is interconnected, players learned that while not all issues can be resolved, prioritizing quick and effective solutions can mitigate broader impacts. Community involvement is essential, particularly in decisions regarding water release during extreme rainfall, ensuring that farmers and local residents are actively consulted.

Leadership and communication also emerged as crucial factors, with village leaders were encouraged to hold regular meetings to promote the King's philosophy and gradually build acceptance within the community. Players recognized the need to balance dialogue and enforcement, where mutual discussions should be prioritized, but legal measures must be in place if agreements fail. The game also highlighted the importance of self-reliance, emphasizing practical solutions like controlling water flow before it reaches critical areas. Farms located outside designated zones should focus on self-sufficiency to reduce dependency on external support. Additionally, targeted erosion control was identified as an efficient strategy, prioritizing high-risk riverbank areas to prevent severe damage and minimize water intensity.

Another key takeaway was the need for greater resource awareness. While people acknowledge the importance of water during shortages, many lack an understanding of its sources and overall significance, highlighting the necessity of education. Effective communication techniques play a vital role in ensuring fair and sustainable resource use. Clear guidelines should be established from the outset to define communal resource usage, preventing exploitation for personal gain. Encouraging fairness and reflection is equally important, as individuals must consider how their actions impact the broader community. Transparent communication methods, such as using bells and notice boards to announce meetings, ensure that communal spaces are managed fairly. Players also emphasized the benefits of collaborating with the agricultural sector, where discussions on food production management and small financial contributions (e.g., 1 THB per person per month) can support maintenance and sustainability. Lastly, simplifying communication can enhance engagement, ensuring that more people understand and participate in decision-making processes. These lessons reinforce the importance of proactive, inclusive, and well-structured approaches to managing natural resources effectively.

Furthermore, this game allows stakeholders—such as community members or local actors with limited authority—to explore complex urban agriculture systems, understand trade-offs, and simulate decision-making without requiring actual control over resources. This approach can foster critical thinking and collaborative learning, ultimately supporting more informed engagement in real-world planning or advocacy processes.

3.6. Exploratory Analysis of Game Outputs and Policy Implications

Local governments should establish a one-stop service to streamline access for residents and improve inter-agency collaboration. Currently, governance is complicated by

overlapping responsibilities among agencies managing agriculture, water, soil, and land titles. Stronger coordination, especially between the Royal Irrigation Department and the Thai Meteorological Department, is essential for adaptive climate planning. The practical policy recommendations are outlined below:

(1) Reforming water resource governance: a centralized coordination mechanism should be established to enhance collaboration among agencies managing land, water, and agriculture. Bureaucratic procedures must be simplified to improve transparency and efficiency in resource allocation. This will help ensure that services reach the intended beneficiaries promptly and without unnecessary delays;
(2) Improving water access and equity: barriers to government support programs, such as the pond construction fund, must be addressed by revising eligibility criteria to be more inclusive, allowing small landholders to benefit, who are often excluded due to land size requirements. Furthermore, preventing the monopolization of water resources is critical. There must be stronger public oversight of privatization efforts to ensure fair access to water resources and avoid concentrating control in the hands of a few entities;
(3) Enhancing infrastructure and maintenance: a dedicated budget should be allocated for the routine maintenance of water systems to prevent breakdowns and reduce long-term repair costs. Authorities should move away from relying on reactive maintenance and implement regular inspections to ensure continuous water access, reducing the risk of disruptions and conflicts over resource allocation;
(4) Leveraging technology for sustainable management: integrating real-time data systems, mobile applications, and online platforms can significantly enhance planning, decision-making, and transparency in water resource management. Additionally, the use of predictive analytics can improve water allocation and disaster preparedness, helping to adapt to climate variability and mitigate the impacts of extreme weather events;
(5) Ensuring long-term resilience and disaster preparedness: a comprehensive funding strategy for disaster preparedness should be developed, ensuring that resources and budgets are available to address potential future floods and droughts. Regular drills and training programs should be conducted to build local capacity for emergency response and climate adaptation, fostering resilience in the face of environmental challenges. By implementing these policy recommendations, local authorities can strengthen governance, improve resource efficiency, and enhance resilience against environmental challenges. These measures will lead to more sustainable and equitable natural resource management, benefiting both communities and ecosystems in the long run.

Furthermore, the game results can inform concrete action guidelines for local governments, which is crucial for maximizing the impact of this study. By facilitating bottom-up inputs from citizens—such as urban farmers and community leaders—the game supports the integration of local perspectives into more context-sensitive and participatory policy design. Its simulated crises and trade-offs (e.g., limited water supply and infrastructure failure) provide a valuable foundation for scenario-based planning, helping municipalities develop adaptive strategies and contingency plans. Additionally, the game serves as a practical stakeholder engagement tool in public consultations and training programs, raising awareness and fostering dialogue among local authorities, residents, and sectoral agencies. Finally, both quantitative and qualitative data collected during and after gameplay—such as decision paths and group discussions—can be analyzed to identify recurring challenges and locally endorsed solutions, which can then be translated into targeted guidelines or policy briefs.

3.7. Recommendations for Future Research

We recommend that future studies adopt longitudinal designs to track participants over time following gameplay, in order to monitor changes in pro-environmental attitudes, knowledge retention, and real-world behaviors. Such research could incorporate follow-up surveys, in-depth interviews, or observational methods conducted over extended periods to evaluate the durability of learning outcomes and behavioral transformations. Additionally, integrating behavioral tracking tools—such as community engagement metrics, self-reported lifestyle changes, or participation in local sustainability initiatives—into future versions or extensions of the game would provide deeper insights into the game's long-term influence. These approaches can enhance our understanding of how interactive educational tools translate into sustained, real-world environmental action.

Although the current game includes cards representing key environmental stressors, it does not yet fully capture the wide array of factors influencing urban agriculture, such as market fluctuations, pest outbreaks, and policy changes. Future research should aim to integrate these additional variables into the gameplay to increase the realism and complexity of the scenarios. Broadening the range of stressors would provide a more comprehensive view of the dynamic challenges urban agricultural communities face. Additionally, future studies should employ a randomized controlled trial or quasi-experimental design to more rigorously validate the game's effectiveness.

4. Conclusions

This study highlights the effectiveness of serious games as an innovative educational tool to deepen understanding of the Water–Energy–Land–Food–People (WELFP) Nexus in the context of urban agriculture. By integrating the Stimulus–Organism–Response (SOR) model and the Easy–Attractive–Social–Timely (EAST) framework with the Revised New Ecological Paradigm (NEP) Scale, the game-based learning model successfully engaged diverse participants and fostered participatory decision-making.

Quantitative results demonstrate that perceived ease of use, timeliness, enjoyment, and usefulness significantly influenced the perceived value of the game, which, together with positive attitudes, strongly predicted participants' behavioral intention to apply the game's lessons in real-life contexts. These findings also align with core constructs from the Technology Acceptance Model (TAM), reinforcing the relevance of game-based interventions in sustainability education.

The perceived value (PV) of the game, influenced by perceived ease of use (PEOU), timeliness (PT), enjoyment (PE), and usefulness (PU), was a strong predictor of participants' intention to apply the game's lessons in real life. PEOU and PT had strong positive effects on PV, while PEOU did not significantly impact attitude. Perceived enjoyment (PE) positively influenced user attitude but had a negative effect on perceived usefulness. Both attitude (ATT) and PV significantly predicted behavioral intention (BI), highlighting the importance of balancing engaging and practical elements to enhance real-life application.

Academically, this study contributes to the growing body of literature on serious games by offering a validated framework that combines behavioral science, environmental education, and technology acceptance models. It advances the understanding of how digital and analog game-based interventions can promote complex systems thinking, especially within sustainability education.

Practically, the NEXUS game was shown to be accessible, engaging, and effective for knowledge transfer, skill development, and attitude change among urban agriculture practitioners. Players gained practical insights into resource coordination, policy trade-offs, crisis management, and adaptive decision-making. The game's collaborative and low-barrier design makes it a viable tool for schools, community centers, and training workshops.

From a policy perspective, the game generated rich qualitative insights during debriefing sessions, emphasizing community-driven approaches, inclusive leadership, and transparent communication as cornerstones of resilient urban agriculture. The policy implications include the need for improved cross-sector coordination, equitable access to resources, proactive maintenance of urban infrastructure, and the integration of digital tools to support adaptive governance.

Overall, the NEXUS game demonstrates strong potential to bridge academic knowledge, practical skills, and participatory policymaking. Future research should explore its long-term behavioral impact, expand its complexity to reflect additional urban agriculture challenges, and investigate strategies to increase its accessibility and inclusivity across diverse communities.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/su17094148/s1: Table S1: Knowledge or assistance cards; Table S2: Heterotrait–monotrait ratio (HTMT)—List; Table S3: Fornell–Larcker criterion; Table S4: Collinearity statistics (VIF); Table S5: The R-square (R^2) results; Figure S1: The role and situation cards; Figure S2: Examples of knowledge cards.

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Abbreviations

Water–Energy–Land–Food–People (WELFP); Stimulus–Organism–Response (SOR); Easy–Attractive–Social–Timely (EAST); New Ecological Paradigm (NEP); perceived ease of use (PEOU); perceived usefulness (PU); perceived enjoyment (PE); perceived practicality (PP); perceived social norm (PSN); perceived timeliness (PT); attitude (ATT); perceived value (PV); behavioral intention (BI); contributing to nature (CTN); contributing to science and practical solutions (CTS); contributing to useful information (CTI); Variance Internal Factor (VIF); and Technology Acceptance Model (TAM).

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